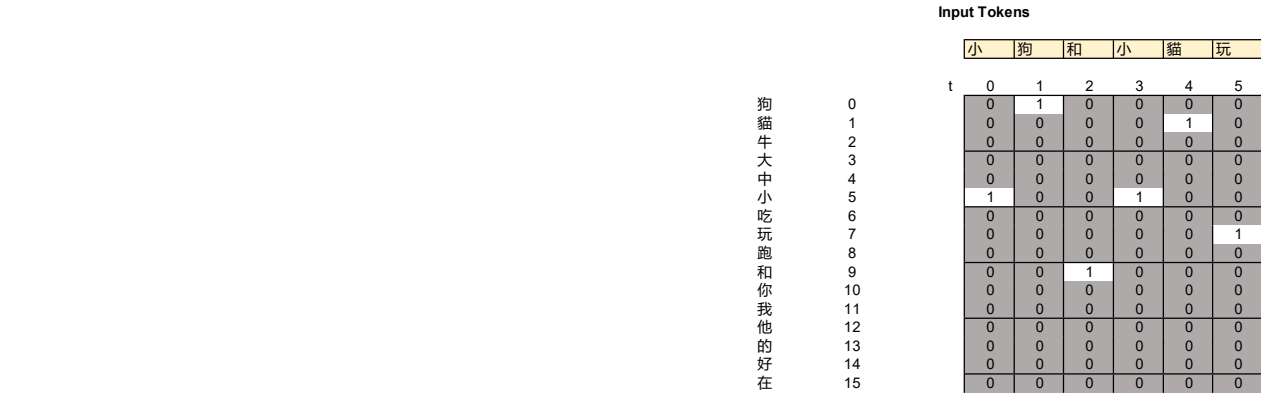
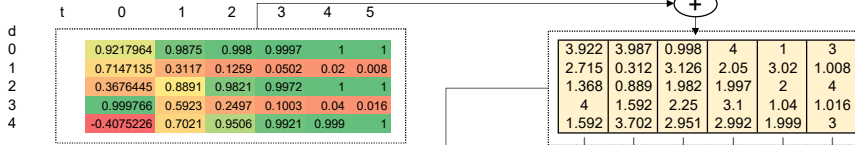




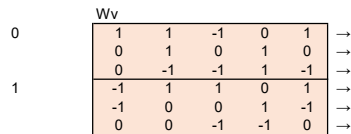
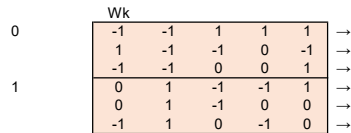
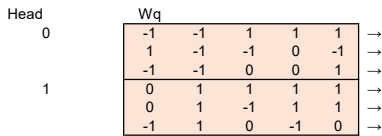
Word Embedding



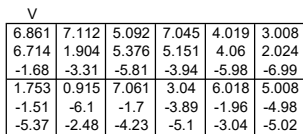
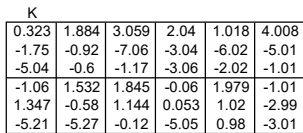
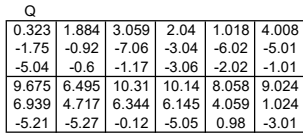
Positional Encoding



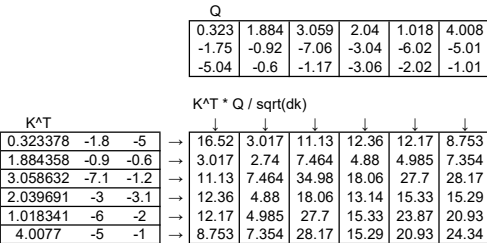
Transformer Encoder



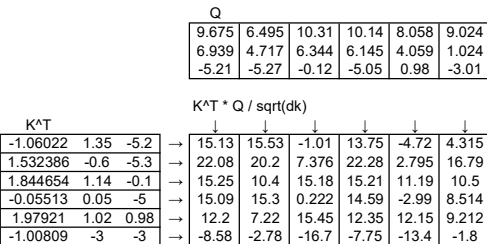
Multi-head Self-Attention



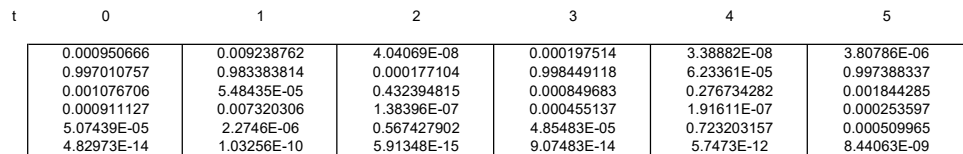
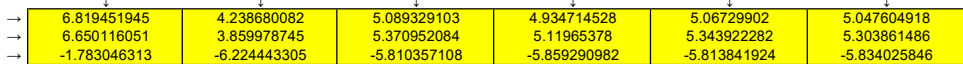
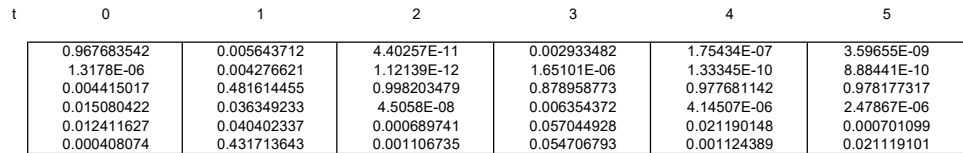
Head 0



Head 1



V



↓

↓

↓

↓

↓

↓

1.753	0.915	7.061	3.04	6.018	5.008
-1.51	-6.1	-1.7	-3.89	-1.96	-4.98
-5.37	-2.48	-4.23	-5.1	-3.04	-5.02

→ 0.925021621	→ 0.939052713	→ 6.468122953	→ 0.922015745	→ 6.306465947	→ 0.929892003
→ -6.085922361	→ -6.038498381	→ -1.847123111	→ -6.091387242	→ -1.887083396	→ -6.086435347
→ -2.48845653	→ -2.52732692	→ -3.554990146	→ -2.48469143	→ -3.369458474	→ -2.485603691

6.819451945	4.238680082	5.089329103	4.934714528	5.06729902	5.047604918
6.650116051	3.859978745	5.370952084	5.11965378	5.343922282	5.303861486
-1.783046313	-6.224443305	-5.810357108	-5.859290982	-5.813841924	-5.834025846
0.925021621	0.939052713	6.468122953	0.922015745	6.306465947	0.929892003
-6.085922361	-6.038498381	-1.847123111	-6.091387242	-1.887083396	-6.086435347
-2.48845653	-2.52732692	-3.554990146	-2.48469143	-3.369458474	-2.485603691

↓ 1.630431839	↓ -2.758063673	↓ 4.21275599	↓ -2.452583806	↓ 3.862082497	↓ -2.418530152
9.014077473	-0.697903228	12.82591407	1.51039726	12.3862204	1.846500905
-9.014077473	0.697903228	-12.82591407	-1.51039726	-12.3862204	-1.846500905
-6.599840541	0.397489016	3.634160813	-0.638099231	3.683550376	-0.769290759
-15.09999983	-5.340595154	-14.67303718	-7.601784502	-14.2733038	-7.932936251

Linear					
0	0	1	1	0	-1
1	1	1	1	1	-1
-1	-1	-1	-1	-1	1
-1	0	-1	1	0	1
-1	-1	-1	-1	0	1

Scale & Shift					
gamma			beta		
1	0	0	0	0	2
0	2	0	0	0	3
0	0	5	0	0	0
0	0	0	3	0	1
0	0	0	0	5	2

Linear					
-1	-1	1	1	-1	0
1	0	1	0	1	0
-1	1	-1	1	1	-1
0	0	0	-1	0	-1
-1	0	0	1	-1	0
0	1	0	1	1	1
1	0	1	1	1	0
-1	0	0	1	0	-1
-1	0	1	-1	0	1
0	0	0	-1	1	-1
1	1	0	0	-1	0
1	0	0	1	-1	0

Linear											
0	1	0	0	0	0	1	-1	1	-1	0	-1
0	-1	0	0	0	0	1	-1	1	0	0	-1
-1	1	0	1	1	-1	0	1	1	-1	0	1
1	1	1	1	-1	1	-1	-1	1	0	0	-1
0	-1	0	1	0	-1	1	1	0	1	0	1

Add & Norm

5.552	1.229	5.211	1.547	4.862	0.581
11.73	-0.39	15.95	3.561	15.41	2.854
-7.65	1.587	-10.8	0.487	-10.4	2.153
-2.6	1.99	5.884	2.462	4.724	0.247
-13.5	-1.64	-11.7	-4.61	-12.3	-4.93

Normalize

0.678	0.442	0.362	0.27	0.378	0.131
1.29	-0.62	1.264	0.905	1.285	0.875
-0.63	0.676	-0.99	-0.06	-0.93	0.645
-0.13	0.941	0.419	0.559	0.366	0.022
-1.21	-1.44	-1.06	-1.67	-1.1	-1.67
1	1	1	1	1	1

↓ 2.678	↓ 2.442	↓ 2.362	↓ 2.27	↓ 2.378	↓ 2.131
5.58	1.763	5.528	4.81	5.57	4.749
-3.15	3.382	-4.93	-0.32	-4.67	3.227
0.612	3.822	2.256	2.676	2.098	1.065
-4.05	-5.2	-3.3	-6.35	-3.48	-6.36
↓	↓	↓	↓	↓	↓

Feed-Forward

↓ -6.74	↓ 8.201	↓ -7.27	↓ 1.627	↓ -7.04	↓ 3.775
-4.52	0.622	-5.86	-4.4	-5.77	-1.01
1.611	-6.44	6.054	-1.82	5.478	-6.91
-1.61	-4.82	-3.26	-3.68	-3.1	-2.06
1.982	6.582	3.192	6.756	3.199	5.298
3.144	1.384	5.487	2.136	5.19	0.45
-3.9	4.444	-3.61	-1.72	-3.67	0.059
-3.07	0.381	-1.11	-0.59	-1.28	-2.07
-5.44	-1.88	-8.55	-4.27	-8.14	1.031
-5.66	-10	-6.55	-10	-6.58	-8.43
12.31	9.406	11.19	13.43	11.43	13.24
7.338	11.47	7.917	11.3	7.955	9.56

ReLU

0	8.201	0	1.627	0	3.775
0	0.622	0	0	0	0
1.611	0	6.054	0	5.478	0
0	0	0	0	0	0
1.982	6.582	3.192	6.756	3.199	5.298
3.144	1.384	5.487	2.136	5.19	0.45
0	4.444	0	0	0	0.059
0	0.381	0	0	0	0
0	0	0	0	0	1.031
0	0	0	0	0	0
12.31	9.406	11.19	13.43	11.43	13.24
7.338	11.47	7.917	11.3	7.955	9.56

↓ -7.34	↓ -6.78	↓ -7.92	↓ -11.3	↓ -7.95	↓ -8.47
-8.34	-9.02	-8.92	-12.3	-8.95	-9.47
6.176	9.466	5.621	14.29	5.964	11.66
-5.57	-13.7	-0.57	-15.3	-1.49	-10.7
3.194	13.28	1.429	8.162	1.765	8.168

ReLU

0	0	0	0	0	0
0	0	0	0	0	0
6.176	9.466	5.621	14.29	5.964	11.66
0	0	0	0	0	0
3.194	13.28	1.429	8.162	1.765	8.168

Add & Norm

2.678	2.442	2.362	2.27	2.378	2.131
5.58	1.763	5.528	4.81	5.57	4.749
3.031	12.85	0.692	13.97	1.297	14.89
0.612	3.822	2.256	2.676	2.098	1.065
-0.85	8.083	-1.87	1.811	-1.71	1.805

Normalize

0.191	-0.72	0.21	-0.56	0.173	-0.49
1.37	-0.87	1.384	-0.06	1.398	-0.03

Scale & Shift

gamma						beta					
2	0	0	0	0	2	2	0	0	0	0	2
0	1	0	0	0	0	3	0	0	0	0	3
0	0	2	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	1	0	0	0	0	1
0	0	0	0	2	2	2	0	0	0	0	2

0.334	1.518	-0.41	1.742	-0.24	1.735
-0.65	-0.42	0.171	-0.48	0.066	-0.67
-1.25	0.493	-1.36	-0.65	-1.4	-0.54
1	1	1	1	1	1

2.381	0.559	2.421	0.885	2.347	1.026
4.37	2.133	4.384	2.941	4.398	2.969
0.668	3.036	-0.82	3.485	-0.48	3.471
0.351	0.576	1.171	0.522	1.066	0.327
-0.49	2.986	-0.71	0.704	-0.79	0.912

